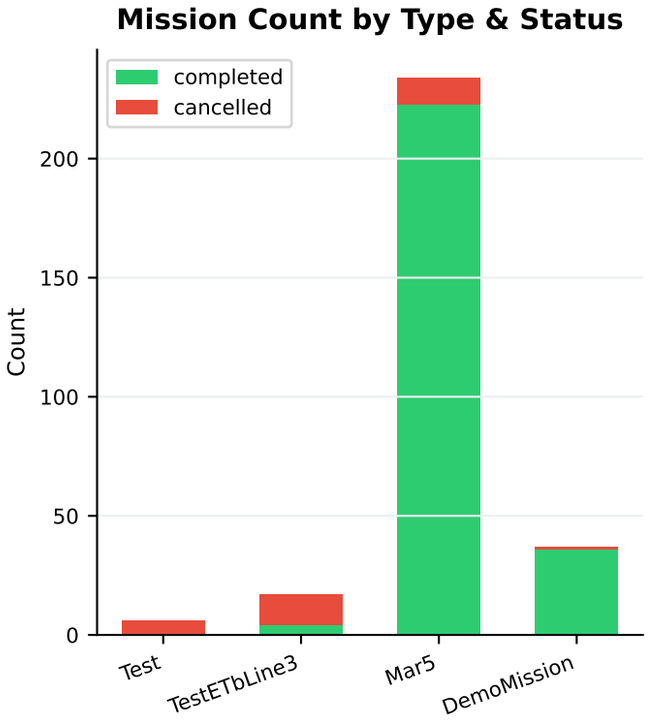
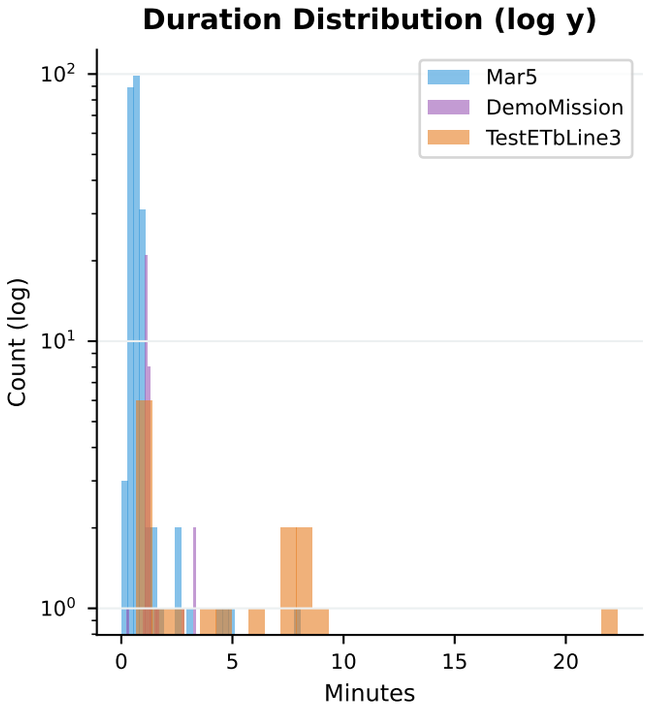
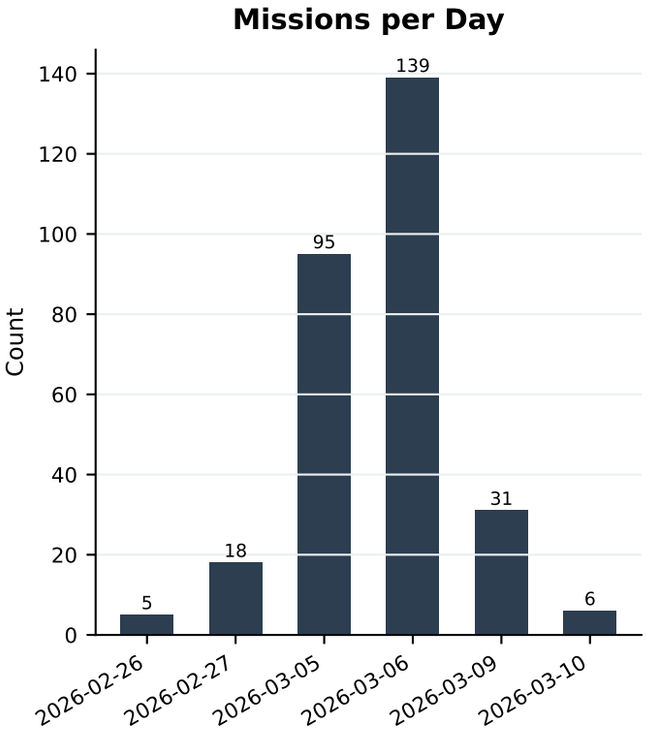
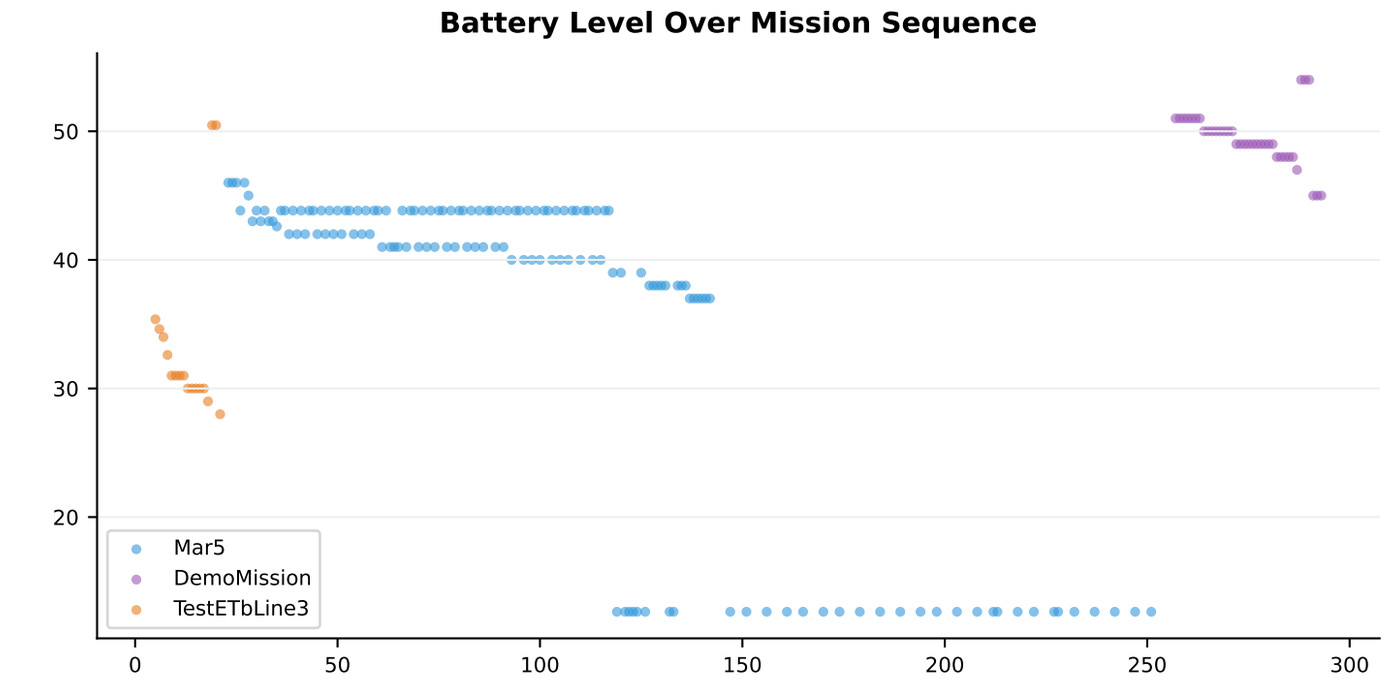
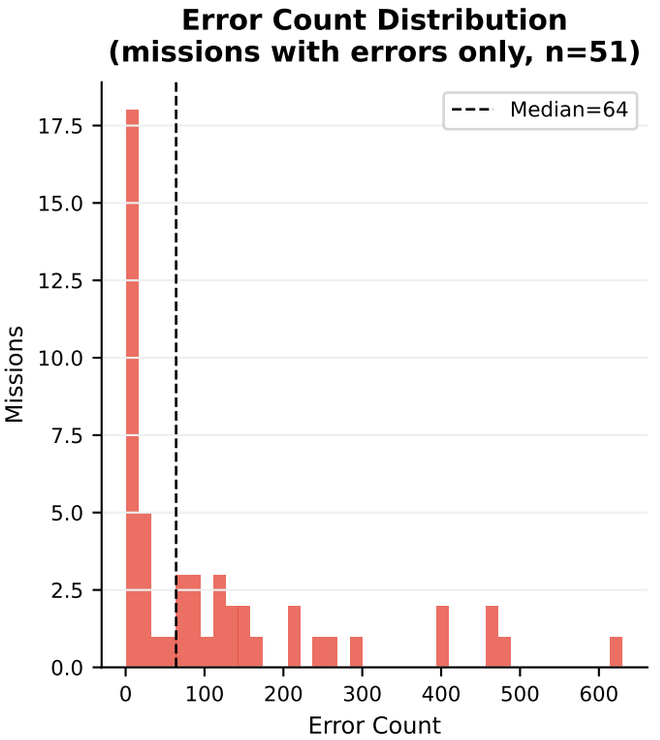
AMR Phase 2 EDA — Dataset Overview





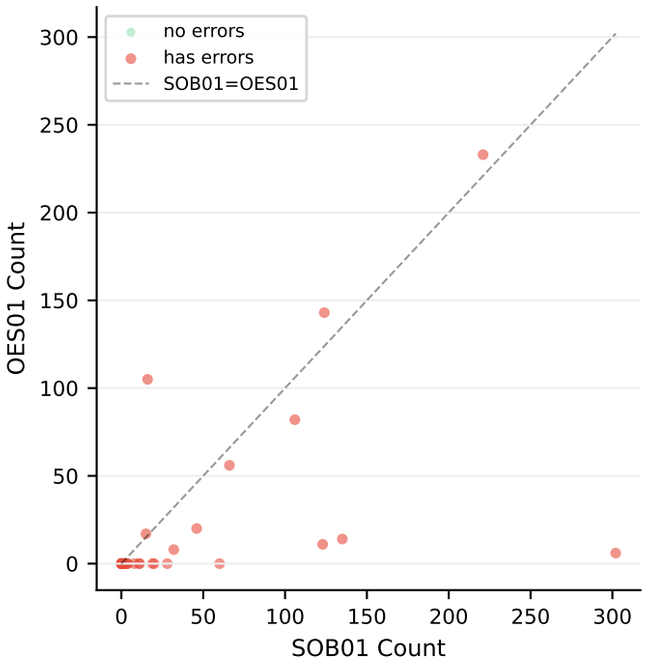




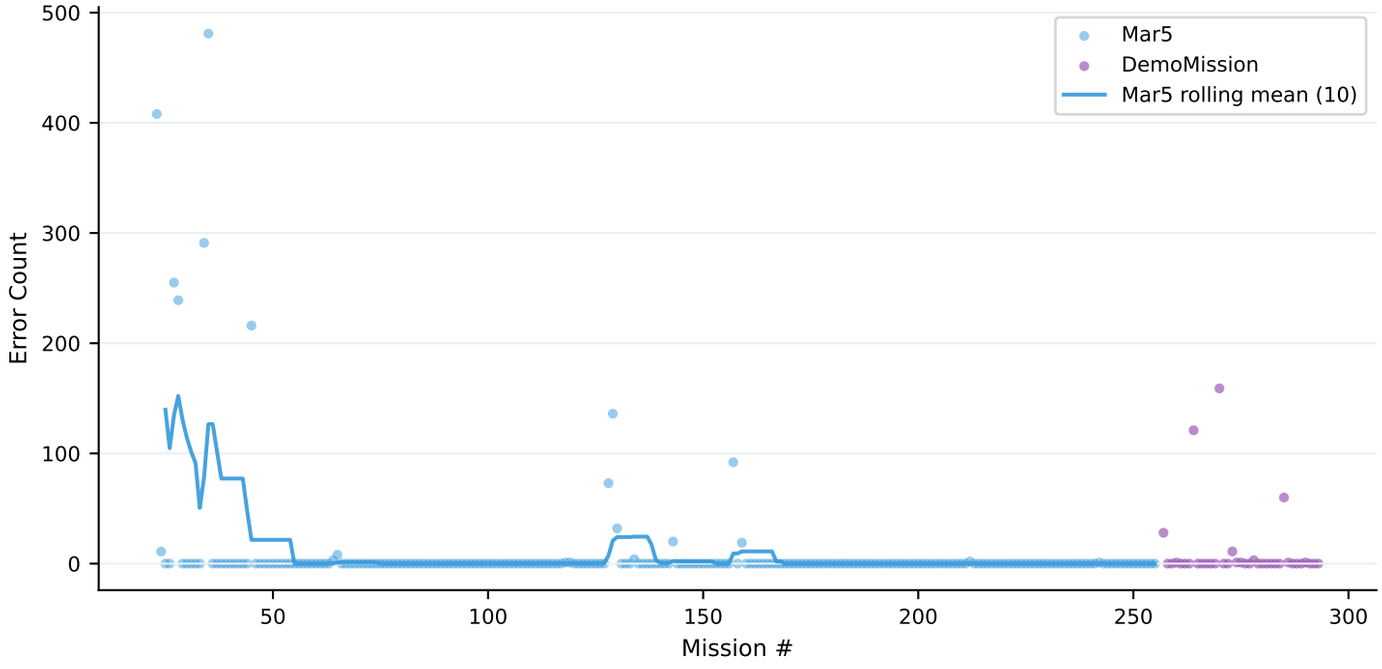


Error Pattern Analysis — AMR12601071239XCR

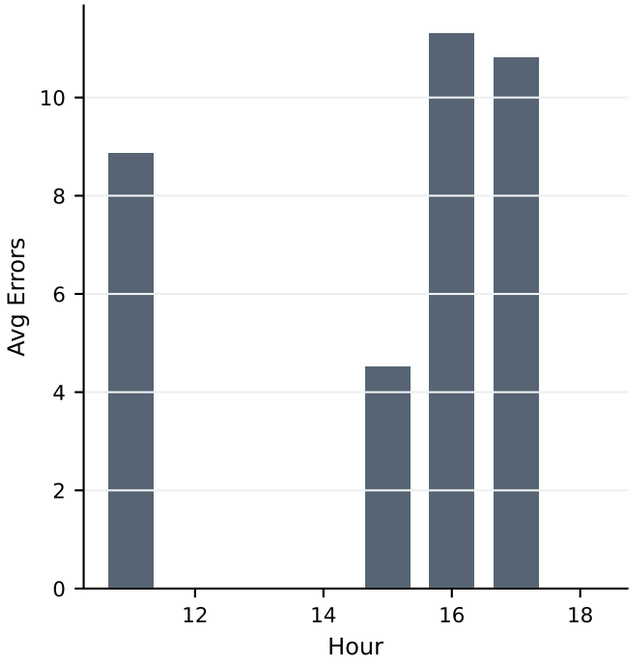
SOB01 vs OES01 Count per Mission



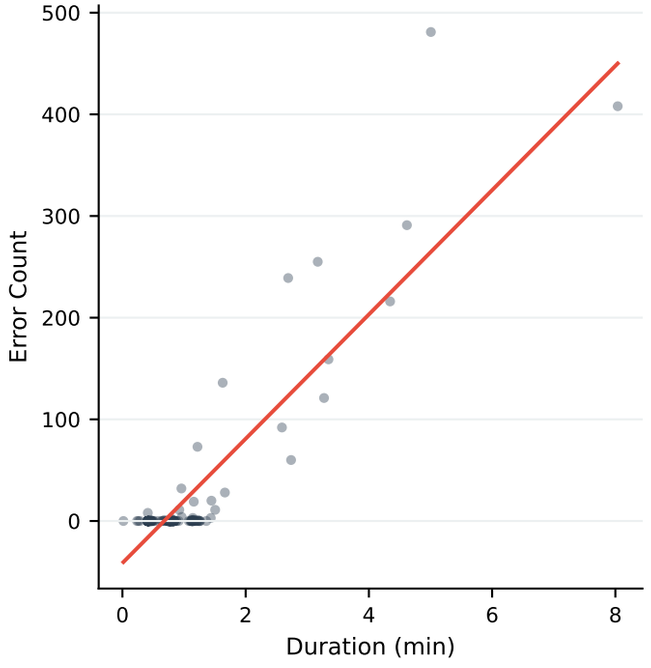
Error Count Over Mission Sequence



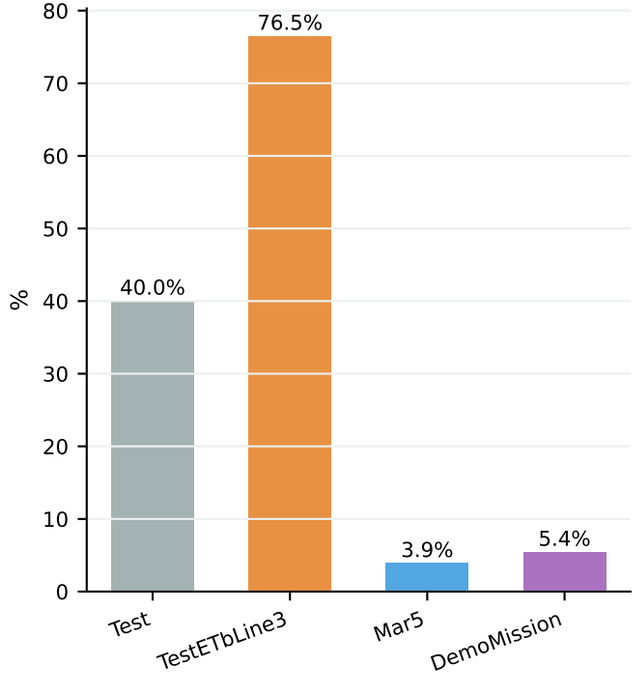
Avg Error Count by Hour of Day (IST)



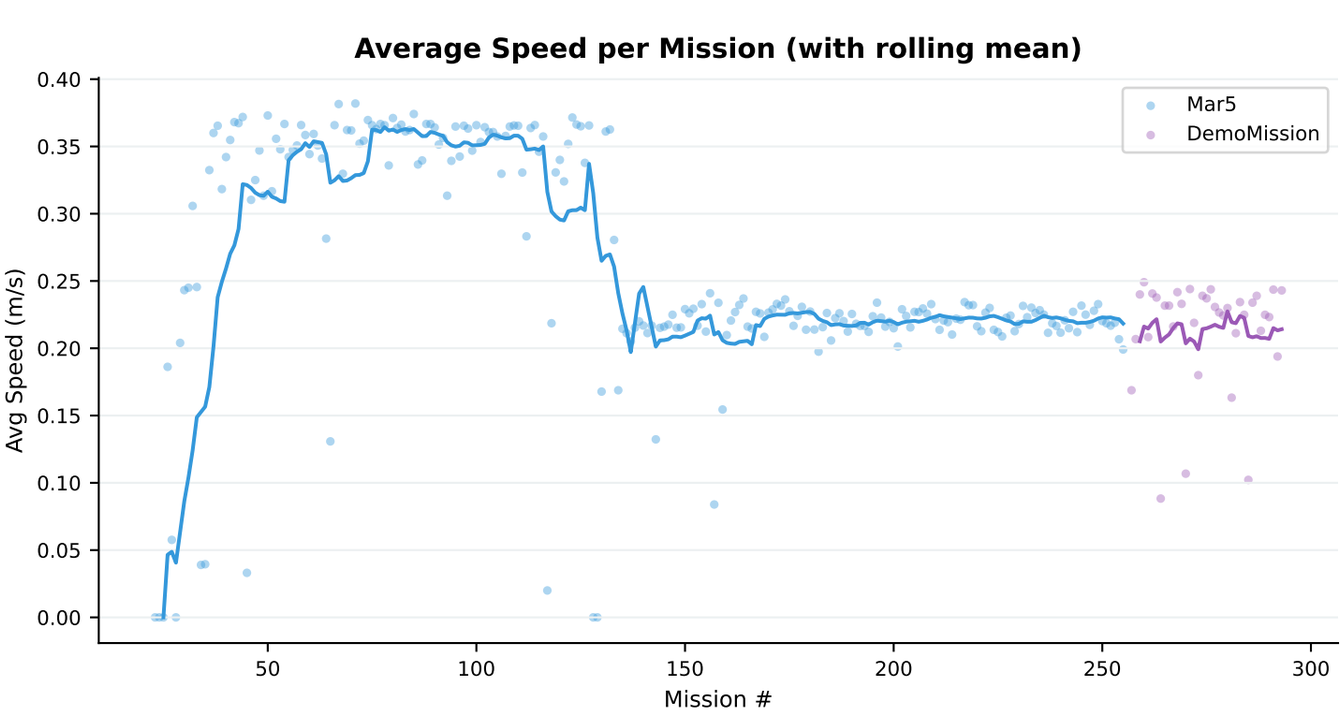
Error Count vs Duration
r=0.88, p=0.000

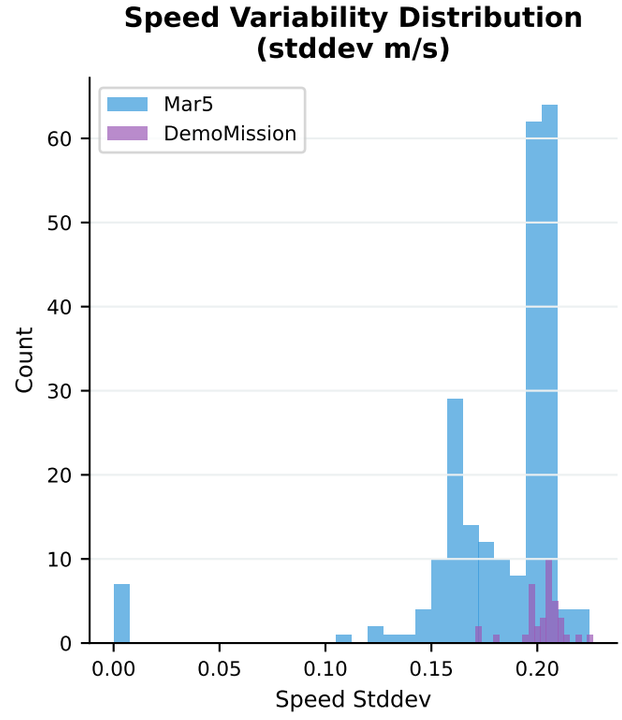


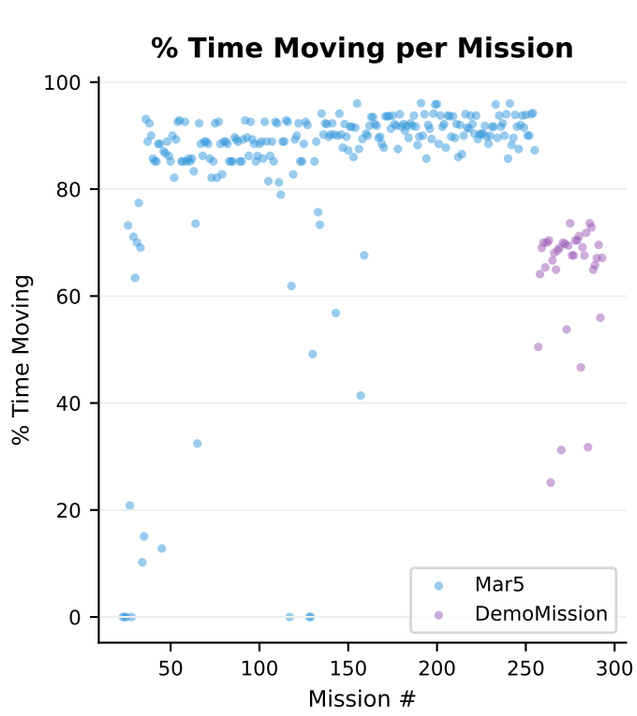
SOB01→OES01 Escalation Rate by Mission Type

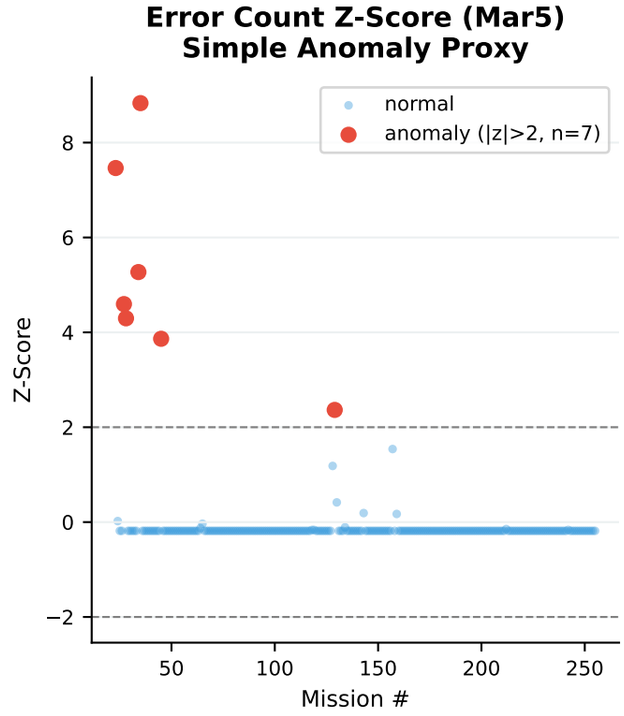


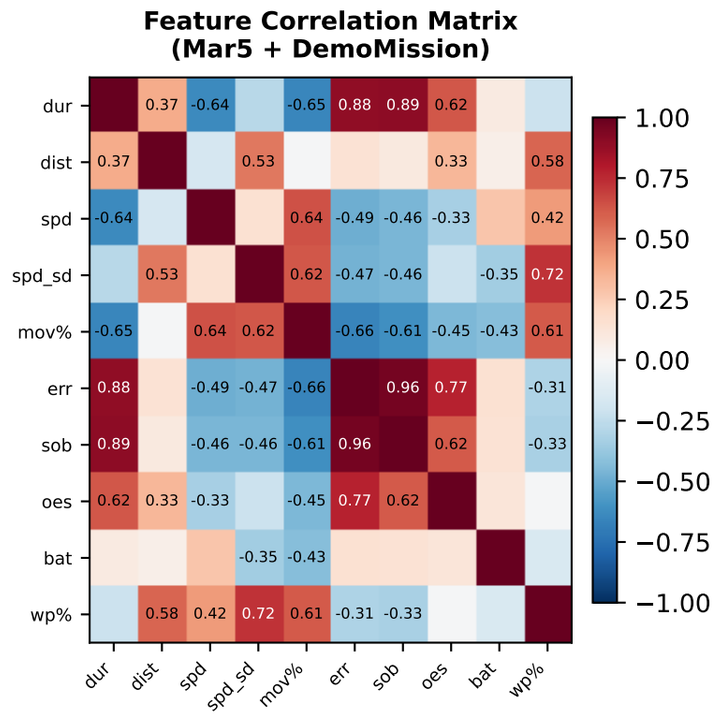
Performance Metrics & Anomaly Signals — AMR12601071239XCR



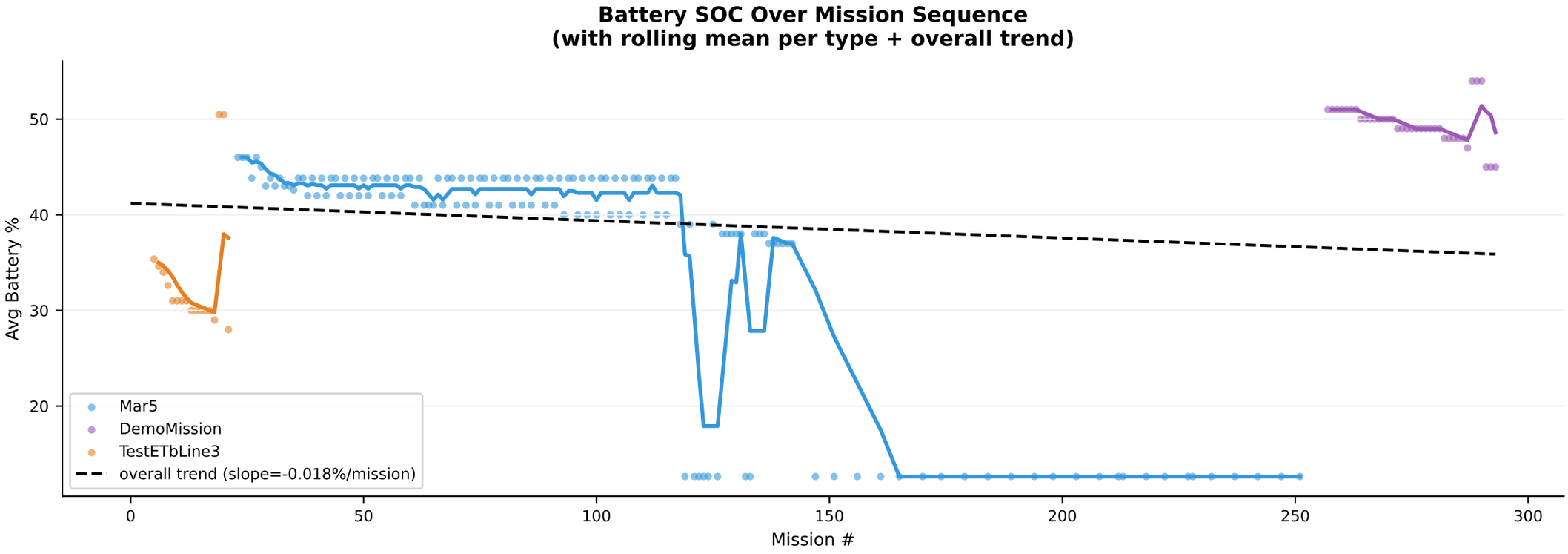


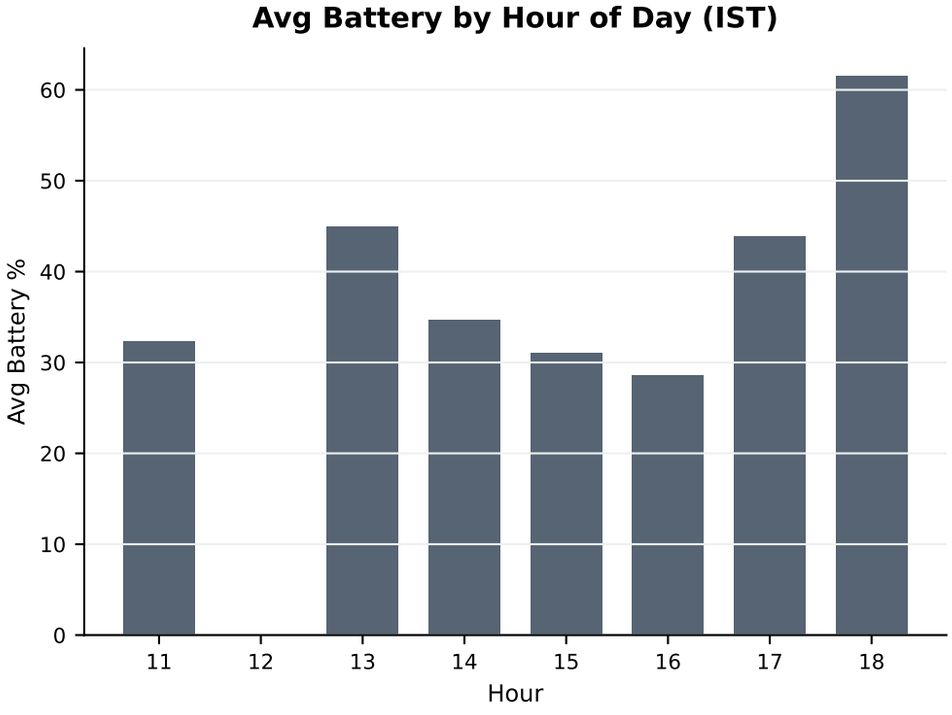


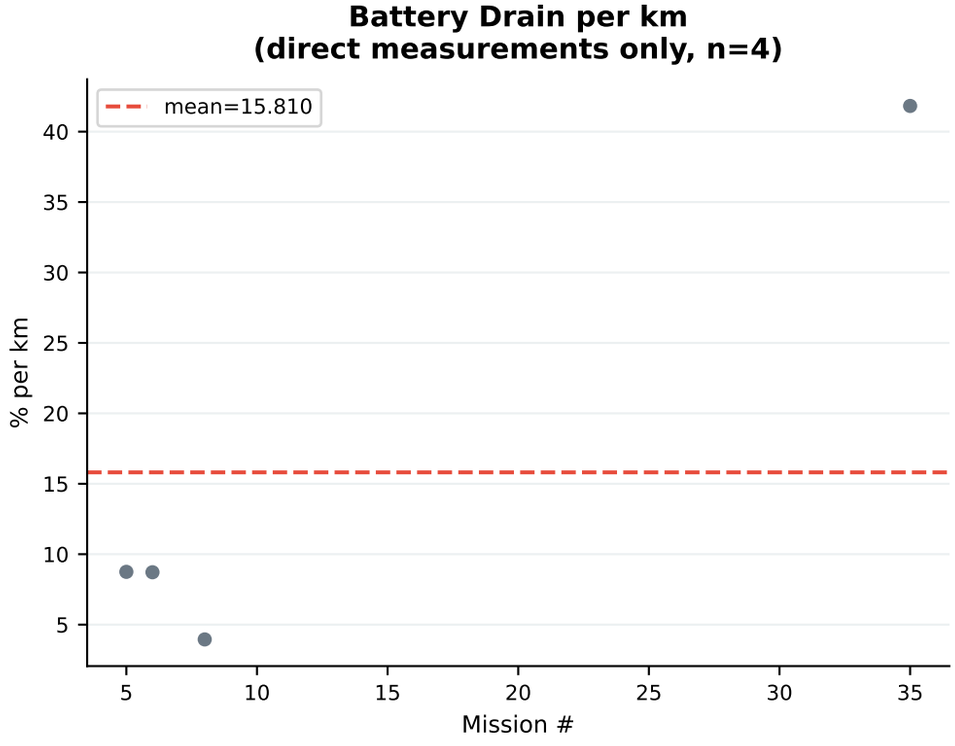




Battery Analysis & Degradation Signals — AMR12601071239XCR







ML Feasibility Assessment

```
DATASET FACTS
──────────────────────────────────────────────────────────────
Robot with usable data : AMR12601071239XCR only
Usable missions        : 294 (after filtering stuck mission)
Date span              : Feb 26 – Mar 10, 2026 (13 days)
Mission types          : Mar5 (234), DemoMission (37), TestETbLine3 (17), Test (6)
Failure labels         : NONE – no component failures, only operational stops

WHAT IS FEASIBLE
──────────────────────────────────────────────────────────────
1. ANOMALY DETECTION (Isolation Forest / Z-score baseline)
   - Features available: error_count, sob01_count, oes01_count, avg_speed_ms,
                         speed_stddev, pct_time_moving, duration_minutes, distance_m
   - Signal confirmed: error count varies 0–6798 per mission with clear outliers
   - Approach: train on "normal" missions (low error, completed), flag deviations
   - Scope: PER MISSION TYPE only (Mar5 and DemoMission have different baselines)

2. OPERATIONAL DISRUPTION CLASSIFIER
   - Target: predict mission cancellation (binary: completed vs cancelled)
   - Cancelled rate: ~10.5% of XCR missions – imbalanced but workable
   - Features: duration, distance, error_count, avg_battery, avg_speed
   - Caveat: most cancellations are early (short duration) – model may just learn duration

3. BATTERY SOC MONITORING (not predictive – tracking only)
   - SOC drains from ~86% to ~45% over 13 days
   - Per-mission battery drain is measurable for direct-source missions only (31 missions)
   - Not enough data for degradation curve fitting

WHAT IS NOT FEASIBLE
──────────────────────────────────────────────────────────────
- True RUL (Remaining Useful Life): no failure labels, no component failure history
- Cross-robot generalization: only 1 active robot, 2 others have trivial data
- Battery degradation modeling: 13 days is too short, rated capacity unknown
- Time-series forecasting: mission timing is irregular, gaps between days

RECOMMENDED NEXT STEP
──────────────────────────────────────────────────────────────
Build an Isolation Forest anomaly detector using Mar5 missions as baseline.
Features: [error_count, sob01_count, avg_speed_ms, speed_stddev,
           pct_time_moving, duration_minutes, waypoint_completion_rate]
Expected output: anomaly score per mission → flag top 5% as "At-Risk"
This is the most honest and defensible model given the available data.
```